ZICHUN YE

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Research Interest

- **Sequential Decision Making:** Online learning, bandits, reinforcement learning.
- **Game Theory and Multi-Agent Learning:** Regret minimization, equilibrium computation.

Education

Shanghai Jiao Tong University **Sep. 2024 - Mar. 2027 (Expected)**
MA.Eng. in Computer Science and Engineering *Shanghai, China*
- GPA 3.87/4.0.

Shanghai Jiao Tong University **Sep. 2020 - Jun. 2024**
B.Eng. in Computer Science and Engineering *Shanghai, China*
- GPA 90.77/100 (or 3.90/4.3).
- Achieved A on more than 30 courses, including all theoretical computer science courses (Discrete Mathematics, Probabilistic Theory, Computing Theory, etc).

Publications

Y. He, **Zichun**. **Ye**, and C. Zhang. Understanding memory-regret trade-off for streaming stochastic multi-armed bandits. In Proceedings of the ACM-SIAM Symposium on Discrete Algorithms (SODA 2025), pages 3450–3485, 2025.

Zichun. **Ye**, R. Wang, X. Liu, and S. Li. Near-optimal regret for efficient stochastic combinatorial semi-bandits. arXiv preprint arXiv:2508.06247, 2025a.

Zichun. **Ye**, C. Zhang, and J. Zhao. Tight gap-dependent memory-regret trade-off for single-pass streaming stochastic multi-armed bandits. In International Computing and Combinatorics Conference (COCOON 2025), pages 209–222, 2025b.

Zichun. **Ye**, R. Wang, X. Wang, X. Liu, S. Li, and M. Hajiesmaili. Unlearning offline stochastic multi-armed bandits. *arXiv preprint arXiv:2605.00638*, 2026.

Research Experience

Swap Regret Minimization and Game Theory **Mar. 2026 – Present**
Advisor: Prof. Haipeng Luo, USC
- Studied linear swap-regret minimization under bandit feedback without an efficient membership oracle for the set Φ of affine self-maps. Building on prior shell-projection methods, identified bandit linear optimization over shrinking convex action sets as a fundamental subproblem.
- Developed two algorithms for bandit linear optimization over shrinking convex sets with known low-dimensional loss subspaces, combining continuous exponential weights with either geometric exploration or Dikin sampling, and established their respective regret bounds.
- Applied the Dikin-sampling algorithm as a blackbox in a Gordon-style reduction, maintaining shrinking outer approximations of Φ to ensure feasible exploration and achieving $\widetilde{O}(d^2\sqrt{T})$ expected linear swap regret against every fixed affine self-map.

Unlearning for Offline Stochastic Multi-Armed Bandits **Oct. 2025 – Jan. 2026**
Advisor: Prof. Xutong Liu, UW Tacoma, Prof. Mengfan Xu, UMass, Prof. Mohammad Hajiesmaili, UMass
- Initiated the study of sample-level (ε, δ)-unlearning for offline stochastic bandits, formalizing privacy guarantees and measuring utility through post-unlearning decision quality.
- Developed adaptive unlearning algorithms based on two complementary mechanisms: downward-shifted Gaussian perturbation and exact rollback. Used stored statistics and privacy parameters gathered in the learning process to select the mechanism with the better post-unlearning suboptimality guarantee.
- Established upper bounds under fixed-sample and distribution-based models, and derived Le Cam-based lower bounds matching these rates up to logarithmic factors for single-source deletion in several regimes; extended upper-bound guarantees to multi-source deletion under the fixed-sample model.
- Experiments on synthetic and Yelp-derived datasets validated the predicted trade-offs; for multi-source deletion, the adaptive algorithm matched the better base mechanism in 15 of 16 hard-case (one of the deletion requests comes from the best arm) configurations.

Near-Optimal Regret for Efficient Stochastic Combinatorial Semi-Bandits **Mar. 2025 – June. 2025**
Advisor: Prof. Xutong Liu, UW, Tacoma, Prof. Shuai Li, SJTU

- Studied the problem of achieving optimal regret for stochastic combinatorial semi-bandits without the extra $\log T$ factor, while keeping the algorithm computationally efficient.
- Designed CMOSS, a MOSS-inspired algorithm which incorporated new confidence bound into CMAB work.
- Proved near-optimal instance-independent regret bound by devising new technical methods and validated by ablation experiments on multiple algorithms.

Memory-Regret Trade-Off for Streaming Stochastic Multi-Armed Bandits **May. 2023 – Mar. 2025**
Advisor: Prof. Chihao Zhang, SJTU

- Characterized the memory–regret trade-off for streaming stochastic multi-armed bandits by introducing a P-pass streaming model with limited arm memory.
- Identified best-arm retention as the fundamental problem underlying regret minimization in streaming bandits, and designed algorithms for different memory regimes with corresponding regret guarantees.
- Proved the first tight minimax regret lower bound via the technique of likelihood, revealing sharp transitions between small-memory and large-memory regimes.
- Extended the setting to the gap-dependent regret and established tight regret bounds in the single pass.

Additional Experience and Awards

Teaching Assistant *Topics in Modern Algorithms* **Fall 2023, 2024**
Advisor: Prof. Chihao Zhang

- The topics include moment method, concentration inequalities, martingales and diffusion model.
- Responsible for designing, assigning, and marking homework.

Bosch Scholarship **Dec. 2025**
Zhiyuan Honor Scholarship (Top 10%) **Dec. 2021, 2022, 2023**
Huawei Scholarship **Oct. 2022**
COSCO First Class Scholarship **Oct. 2021**

Technical Skills

Programming Languages	Python, C++, LaTeX
Language	TOEFL 5.5 (107+/120, Reading 6, Listening 5.5, Speaking 5, Writing 5.5) GRE (157 VR 170 QR 4.0 AW)